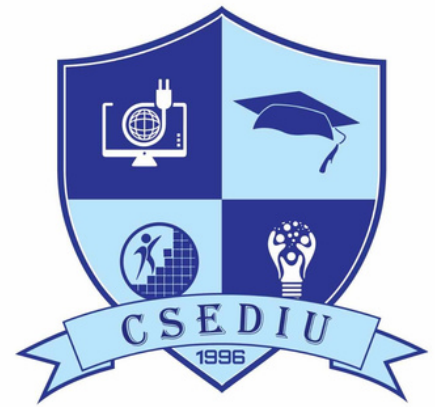




FACULTY OF SCIENCE AND ENGINEERING
DEPARTMENT OF CSE

LAB REPORT

Course Name : Operating System

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Lab Number : 02

Lab Name : Linux Command with Variable Declaration

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Submitted to
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Title: Linux Command with Variable Declaration

Objective:

- To understand variable declaration in Bash scripting.
- To perform arithmetic operations using Bash arithmetic expression
- To develop a simple command-line calculator in Linux.

Introduction:

Bash is a default command-line interpreter in most Linux distributions. It allows users to automate tasks by writing shell scripts. Variables in Bash are used to store values such as numbers, strings, and command outputs. User input can be accepted through the 'read' command, and arithmetic operations can be performed using arithmetic expansions '\$(())'.

In this lab report, a simple calculator program was developed using Bash scripting. The program accepts two numbers and an arithmetic operator from the user, performs selected operation and displays results.

Code:

```
#!/bin/bash
echo "Enter first number"
read num1

echo "choose an operation (+, -, *, /, %):"
read operator

echo "Enter second number"
read num2

case $operator in
    +)
        result=$((num1 + num2))
        ;;
    -)
        result=$((num1 - num2))
        ;;
```

*)

```
result=$(( num1 * num2 ))  
;;
```

/)

```
if [ "$num2" -eq 0 ]; then  
    echo "Error: Division by zero is not allowed."  
    exit 1  
fi
```

```
result=$(( num1 / num2 ))  
;;
```

%)

```
if [ "$num2" -eq 0 ]; then  
    echo "Error: Modulus by zero is not allowed."  
    exit 1  
fi
```

```
result=$(( num1 % num2 ))  
;;
```

*)

```
echo "Error: Invalid operator entered."  
exit 1  
;;
```

esac

```
echo "Results: $num1 $operator $num2 = $result"
```

Output:

```
USER@rexys MINGW64 /f/DIU/5th_Semester/OS/Lab Report/Lab Report 02
$ bash report2.sh
===== Basic Calculator =====
Enter first number:
12
Choose an operation (+, -, *, /, %):
+
Enter second number:
13
Result: 12 + 13 = 25

USER@rexys MINGW64 /f/DIU/5th_Semester/OS/Lab Report/Lab Report 02
$ bash report2.sh
===== Basic Calculator =====
Enter first number:
23
Choose an operation (+, -, *, /, %):
-
Enter second number:
12
Result: 23 - 12 = 11

USER@rexys MINGW64 /f/DIU/5th_Semester/OS/Lab Report/Lab Report 02
$ bash report2.sh
===== Basic Calculator =====
Enter first number:
12
Choose an operation (+, -, *, /, %):
*
Enter second number:
13
Result: 12 * 13 = 156

USER@rexys MINGW64 /f/DIU/5th_Semester/OS/Lab Report/Lab Report 02
$ bash report2.sh
===== Basic Calculator =====
Enter first number:
24
Choose an operation (+, -, *, /, %):
/
Enter second number:
3
Result: 24 / 3 = 8

USER@rexys MINGW64 /f/DIU/5th_Semester/OS/Lab Report/Lab Report 02
$ bash report2.sh
===== Basic Calculator =====
Enter first number:
24
Choose an operation (+, -, *, /, %):
/
Enter second number:
0
Error: Division by zero is not allowed.

USER@rexys MINGW64 /f/DIU/5th_Semester/OS/Lab Report/Lab Report 02
$ bash report2.sh
===== Basic Calculator =====
Enter first number:
12
Choose an operation (+, -, *, /, %):
%
Enter second number:
10
Result: 12 % 10 = 2

USER@rexys MINGW64 /f/DIU/5th_Semester/OS/Lab Report/Lab Report 02
$ bash report2.sh
===== Basic Calculator =====
Enter first number:
12
Choose an operation (+, -, *, /, %):
#
Enter second number:
12
Error: Invalid operator entered.
```

Conclusion:

This lab report successfully demonstrated how to declare variables, read user input, perform arithmetic operations and implement conditional execution using the case statement in Bash scripting. The objective of understanding variable declaration and developing a simple Linux command line calculator were achieved. This experiment provides a strong foundation for learning shell scripting, automation and command line programming in Linux.

References:

- The Linux command line
- GNU Bash reference manual